

Marium-E- Jannat

PhD in Computer Science (Human-Computer Interaction)

University of British Columbia – Okanagan

Kelowna, BC, Canada

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Human-Computer Interaction researcher specializing in mobile and wearable interfaces, augmented and virtual reality, embodied interaction, and context-aware interface design. My work designs and examines how people interact with mobile and wearable interfaces across different social settings and usage contexts, with an emphasis on adaptive, efficient, and context-aware interactive systems. I have published at ACM CHI (Best Paper Honorable Mention, 2025), IEEE ISMAR, IEEE TVCG and the International Journal of Human-Computer Studies; mentored undergraduate students and international research interns whose projects led to publications and submissions; and taught computer science at university level for about ten years in Canada and Bangladesh. Prior research experience also includes artificial intelligence, machine learning, and natural language processing.

Research interests: augmented and virtual reality; mobile and wearable interfaces; ubiquitous and context-aware computing; body-anchored and spatial interaction; user performance and experience.

Research Methods: controlled experiments, usability studies, interviews and questionnaires, mixed-methods research, quantitative and qualitative analysis, interactive prototyping, iterative user-centered design.

Google Scholar (September 2026): 455 citations, h-index 12, i10-index 14.

Education

PhD in Computer Science	2022–2026
University of British Columbia – Okanagan, Canada	
Dissertation: An Exploration of Body-Anchored User Interfaces for Augmented Reality Across Different Mobility and Social Contexts	
Impact: Six empirical studies reported in three lead-author publications	
Supervisor: Dr. Mohammad Khalad Hasan	
MSc in Computer Science (transferred to PhD)	2021–2022
University of British Columbia – Okanagan, Canada	
Research topic: Around-device input on commodity smartphones and smartwatches.	
Impact: Two lead-author publications	
Supervisor: Dr. Mohammad Khalad Hasan	
BSc (Engg.) in Computer Science & Engineering	2008–2012
Shahjalal University of Science & Technology, Bangladesh	

Research Experience

Research Assistant	2026–present
Okanagan Visualization and Interaction (OVI) Lab	
University of British Columbia – Okanagan	
Advisor: Dr. Mohammad Khalad Hasan	
Conducting HCI research on mobile, wearable and extended-reality interfaces; designing and running user studies; analyzing quantitative and qualitative data; preparing publications; mentoring students and research interns	

Graduate Research Assistant

2021–2025

University of British Columbia – Okanagan

Advisor: Dr. Mohammad Khalad Hasan

- Developed research plans and interactive prototypes for mobile, wearable and AR/VR systems; conducted controlled user studies; published at CHI, ISMAR, IEEE TVCG and IJHCS; mentored undergraduate students and interns.

Graduate Research Assistant

2023–2024

UBC Integrated Transportation Research (UiTR) Lab and OVI Lab collaboration

Advisors: Dr. Mahmudur Fatmi and Dr. Mohammad Khalad Hasan

- Project on digital technology, travel behavior and smartphone use. Contributed background research, iOS and Android prototypes, ethics applications, data analysis, reporting and manuscript preparation.

MITACS Accelerate Research Intern

2022

University of British Columbia – Okanagan

Advisors: Dr. Apurva Narayan and Dr. Mohammad Khalad Hasan

- Conducted user studies and analyzed and reported findings

Peer-Reviewed Publications

Conference papers

[C6] Marium-E- Jannat, Shukan Miteshkumar Shah, Khalad Hasan. 2025. Exploring Body-Anchored Augmented Reality Interfaces Across Different Mobility and Social Contexts. IEEE International Symposium on Mixed and Augmented Reality (ISMAR 2025), Daejeon, South Korea. [doi:10.1109/ISMAR67309.2025.00147](https://doi.org/10.1109/ISMAR67309.2025.00147)

[C5] Marium-E- Jannat, Ngan Phan, Anmol Chadha, Keiko Katsuragawa, Khalad Hasan. 2025. HAI-AR: Exploring Hand-Anchored Interfaces in Augmented Reality while Walking. Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI 2025), Yokohama, Japan. **Best Paper Honorable Mention Award.** [doi:10.1145/3706598.3713858](https://doi.org/10.1145/3706598.3713858)

[C4] Marium-E- Jannat, Prateek Dhaka, Keiko Katsuragawa, Khalad Hasan. 2024. Exploring Augmented Reality User Interface Transitions Across Mid-Air, On-Body and Physical Surfaces. IEEE ISMAR 2024, Bellevue, WA, USA. [doi:10.1109/ISMAR62088.2024.00134](https://doi.org/10.1109/ISMAR62088.2024.00134)

[C3] Marium-E- Jannat, Khalad Hasan. 2023. Exploring the Effects of Virtually-Augmented Display Sizes on Users' Spatial Memory in Smartwatches. IEEE ISMAR 2023, Sydney, Australia. [doi:10.1109/ISMAR59233.2023.00070](https://doi.org/10.1109/ISMAR59233.2023.00070)

[C2] Thuan Vo, Marium-E- Jannat, Khalad Hasan, David Ahlström. 2022. FaceUI: Leveraging Front-Facing Camera Input to Access Mid-Air Spatial Interfaces on Smartphones. Graphics Interface 2022 (GI 2022), Montréal, Canada. [doi:10.20380/GI2022.06](https://doi.org/10.20380/GI2022.06)

[C1] Marium-E- Jannat, Thuan T. Vo, Khalad Hasan. 2022. Face-Centered Spatial User Interfaces on Smartwatches. Extended Abstracts of the 2022 CHI Conference on Human Factors in Computing Systems (CHI EA '22), New Orleans, USA. [doi:10.1145/3491101.3519720](https://doi.org/10.1145/3491101.3519720)

Journal articles

[J2] Nelusha Nugegoda, Marium-E- Jannat, Khalad Hasan, Patricia Lasserre. 2024. Exploring the Effect of Viewing Attributes of Mobile AR Interfaces on Remote Collaborative and Competitive Tasks. IEEE Transactions on Visualization and Computer Graphics, 30(11), 7288–7298. [doi:10.1109/TVCG.2024.3456177](https://doi.org/10.1109/TVCG.2024.3456177)

[J1] Marium-E- Jannat, Xing-Dong Yang, Khalad Hasan. 2023. Around-device finger input on commodity smartwatches with learning guidance through discoverability. International Journal of Human-Computer Studies, 179, 103105. [doi:10.1016/j.ijhcs.2023.103105](https://doi.org/10.1016/j.ijhcs.2023.103105)

Grant Writing Experience

Michael Smith Health Research BC	2025
Advancing the Use of Evidence in Seniors' Home and Community Care Program	
Project: There's No Place Like Home: A Virtual Reality – Digital Health Intervention to Promote Home Dialysis Transitions.	
Partners: University of British Columbia and Interior Health.	
Role: wrote core proposal sections on VR module implementation, user-centered design, iterative methodology, prototyping and iterative usability testing.	

Teaching Experience

Teaching Assistant	2021–2025
Department of Computer Science, Mathematics, Physics and Statistics	
University of British Columbia – Okanagan, Canada	
Courses: Computer Programming, Advanced Human-Computer, Web Programming, Human-Computer Interaction, Data Structures	
Assistant Professor / Lecturer	2015-2021
Department of Computer Science & Engineering	
Shahjalal University of Science & Technology, Bangladesh	
Courses: Artificial Intelligence, Machine Learning, Computer Networking, Python Programming, Web Programming and Engineering, Database Management Systems, Computer Graphics and Image Processing, Supervised final-year project courses	
Cisco Certified Network Associate (CCNA) Instructor	2016-2020
Cisco Networking Academy Program	
Shahjalal University of Science & Technology	
Lecturer	2012-2015
Department of Computer Science & Engineering	
Sylhet International University, Bangladesh · 2012–2015	
Courses: Artificial Intelligence, Computer Networking, Computer Graphics, Database Management Systems, C Programming; Digital Logic Design	

Student Mentoring and Supervision

Co-mentored with Dr. Mohammad Khalad Hasan (2021-2026)

Raja Ubaidullah (Hamburg University of Technology, Germany) and	2026
Xiaomeng Wang (Tongji University, China)	
Research interns	
Project: Posture-aware interaction intent in augmented reality	
Led to two CHI 2027 submissions	
Lucas Sébastien Ziegler (CESI Graduate School of Engineering, France), Research intern	2025
Project: AR instruction placement for mobile tasks	
Led to one TOCHI submission	

Anmol Chadha (MNIT Jaipur, India), Research Intern and Ngan Phan (BSc, UBC Okanagan), Undergraduate student Project: Body-anchored interfaces in augmented reality Led to a CHI 2025 publication [C5]	2024
Prateek Dhaka (IIT Kharagpur, India), Research intern Project: Physical surface-anchored interfaces in augmented reality Led to an ISMAR 2024 publication [C4]	2023
Thuan Vo (BSc, UBC Okanagan), Work-Study research project Project: Camera Input to Access Mid-Air Spatial Interfaces on Smartphones Work led to publications at GI 2022 [C2]	2021-2022
About 25 undergraduate thesis students Shahjalal University of Science & Technology Projects in AI, machine learning and natural language processing Led to outcomes including multiple conference and journal publications	2015-2021

Presentations and Talks

HAI-AR: Exploring Hand-Anchored Interfaces in Augmented Reality while Walking Yokohama, Japan	CHI 2025
Exploring Augmented Reality User Interface Transitions Across Mid-Air, On-Body and Physical Surfaces Bellevue, WA, USA	ISMAR 2024
Exploring the Effects of Virtually-Augmented Display Sizes on Users' Spatial Memory in Smartwatches Sydney, Australia:	ISMAR 2023
Face-Centered Spatial User Interfaces on Smartwatches New Orleans, USA	CHI 2022

Honors, Awards and Scholarships

Special Recognition for Review ACM CHI	2026
Graduate Dean's Thesis Fellowship (\$6,000) University of British Columbia – Okanagan	2026
Best Paper Honorable Mention Award ACM CHI	2025
Recognition for Highly Useful Review ACM MobileHCI	2025
University Graduate Research Fellowship (\$6,000/year) University of British Columbia – Okanagan	2021-2025
International Four-Year Doctoral Partial Tuition Award (\$3,900/year) University of British Columbia – Okanagan	2022-2025

Graduate Student Travel Grant (\$1,000) University of British Columbia – Okanagan	2023
Special University of BC Okanagan Graduate Award University of British Columbia – Okanagan (\$1,000)	2022
Dean's Entrance Scholarship (\$5,000) University of British Columbia – Okanagan	2021
CSE Departmental Merit Scholarship Shahjalal University of Science & Technology	2028-2012
Government Merit Scholarship Ministry of Education, Government of Bangladesh	2006-2012

Professional Services

Program committee member International Conference on Bangla Speech and Language Processing	2017-2019
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Reviewer

International Journal of Human-Computer Interaction (IJHCI)	2026
ACM Conference on Human Factors in Computing Systems (CHI)	2021-present
ACM Conference on Intelligent User Interface (IUI)	2022-present
ACM Conference on Human-Computer Interaction with Mobile Devices and Services (MobileHCI)	2022-present
ACM Conference on Designing Interactive Systems (DIS)	2022-present
ACM SIGCHI Conference on Computer-Supported Cooperative Work & Social Computing (CSCW)	2023-present
ACM Symposium on Virtual Reality Software and Technology (VRST)	2023
ACM Conference on Mobile and Ubiquitous Multimedia (MUM)	2023, 2025-26
ACM International Conference on Multimedia Retrieval (ICMR)	2026
ACM Symposium on User Interface Software & Technology (UIST)	2021-2022
IEEE Conference on Virtual Reality and 3D User Interfaces (IEEE VR)	2022-present
IEEE International Symposium on Mixed and Augmented Reality (ISMAR) 2022 – Present	2022-present
IEEE International Conference on 4th Industrial Revolution and Beyond (IC4IR)	2021
IEEE International Conference on Computer and Information Technology (ICCIT)	2016-2020

Additional Information

Work authorization: Permanent resident of Canada

References

1. **Dr. Mohammad Khalad Hasan**
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4. Dr. David Ahlström

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